An Exploratory Evaluation of Mask Fusion and Boundary Refinement for SAR Ship Instance Segmentation

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***Abstract*—Synthetic aperture radar (SAR) ship instance segmentation remains difficult in crowded harbors, where overlapping detections, coarse contours, and unreliable confidence ordering can persist after model training. This article studies a postprocessing cascade for Mask2Former-R50 carriers initialized from Common Objects in Context (COCO) weights and frozen after fitting. The HRSID carrier uses auxiliary mask-quality supervision; the PSeg-SSDD carrier uses the standard head. Adopted weight averaging and multiscale flip test-time augmentation supply aligned candidates. A cluster-support-gated fusion operator groups them by mask intersection-over-union and applies pixel voting only within a bounded cluster-size range; other groups retain their highest-scoring seed. Support denotes candidate cardinality, not the number of distinct views. A separately trained boundary-patch refiner updates only a morphological boundary band, and a learned quality regressor adjusts scores without changing masks. Both correctors use predictions and existing labels from the training split. In the historical HRSID protocol, the selected cascade changes official mask average precision from 45.6 to 59.4 and inshore mask average precision from 27.7 to 44.5. On PSeg-SSDD, a newly trained carrier and dataset-specific correctors yield a change from 60.9 to 67.5 official mask average precision while reusing selected processing settings. An exact replay of the HRSID eight-view candidates finds that per-view top-1 voting exceeds the gated fusion by 2.6 official mask AP points without changing candidates or scores; after applying the stored boundary refiner, the difference is 0.25 points in favor of top-1. These observations are exploratory: HRSID test subsets informed configuration selection, the stronger simple fusion has not been carried through a refitted full cascade, and neither study establishes an acquisition-disjoint advantage.**

***Index Terms*—Boundary refinement, instance segmentation, mask fusion, quality estimation, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) supports maritime observation under cloud cover and at night. Instance segmentation adds vessel contours to detection, providing spatial information for applications such as harbor monitoring and vessel-geometry analysis. Pixel-annotated datasets, including HRSID and PSeg-SSDD, enable evaluation of this task across offshore and inshore scenes [1], [2]. In crowded harbors, bright shore structures, adjacent ships, and weak hull contrast make instance separation and accurate delineation particularly challenging. These conditions also expose an inference problem: several plausible predictions may describe the same vessel, yet their outlines and confidence scores need not agree.

Recent SAR segmentation methods address these difficulties through specialized representations and decoders. Examples include low-level feature guidance [3], orientation-aware queries [4], diffusion-based instance decoding [5], Gaussian-heatmap instance encoding [6], and adaptation of foundation segmentation models [7]. These directions concern how a model learns and represents targets. Once a model has been fitted, a complementary question remains: how should its overlapping predictions be associated, converted into one mask per instance, and ranked? This question is relevant when multiple transformed inputs are evaluated with an existing segmenter and retraining the main network is not part of the inference procedure.

Three aspects of the output motivate the design studied here. First, the candidate set must suppress duplicates while retaining neighboring vessels. Bounding boxes can overlap substantially even when elongated ship masks occupy different pixels, making mask intersection-over-union (IoU) a useful association criterion. Second, combining candidates can change a contour: averaging or voting may reconcile a disagreement, but it can also replace an adequate seed boundary with a poorer one. Third, a retained mask may still have local boundary errors or a confidence score that does not reflect its spatial quality. These are distinct intervention points, although their effects on average precision (AP) remain coupled through matching and ranking.

We address them with a postprocessing cascade around a fixed, already trained carrier (Fig. 1). The HRSID carrier is a Mask2Former with a ResNet-50 backbone [8], initialized from Common Objects in Context (COCO) instance segmentation weights and fitted with auxiliary mask-quality supervision. The PSeg-SSDD carrier instead uses the standard Mask2Former head. Each is frozen after training, and these training identities are part of the respective protocols. Weight averaging following model soups [9] and multiscale horizontal-flip test-time augmentation (TTA) are adopted to generate candidates. Our fusion operator processes candidates in score order, groups unused masks by their IoU with a seed, and uses the number of candidates in a cluster to select its output rule. A bounded support range activates pixel voting, with foreground retained at ties; clusters outside that range keep their seed. The support variable counts candidates and can include multiple predictions from one view. The gate is therefore a cluster-cardinality heuristic, rather than a measure of independent cross-view agreement or a guarantee of mask quality.

Figure 1 summarizes the pipeline. Two supervised correctors then operate on the fused instances. A boundary-patch refiner adopts the local re-segmentation principle of BPR

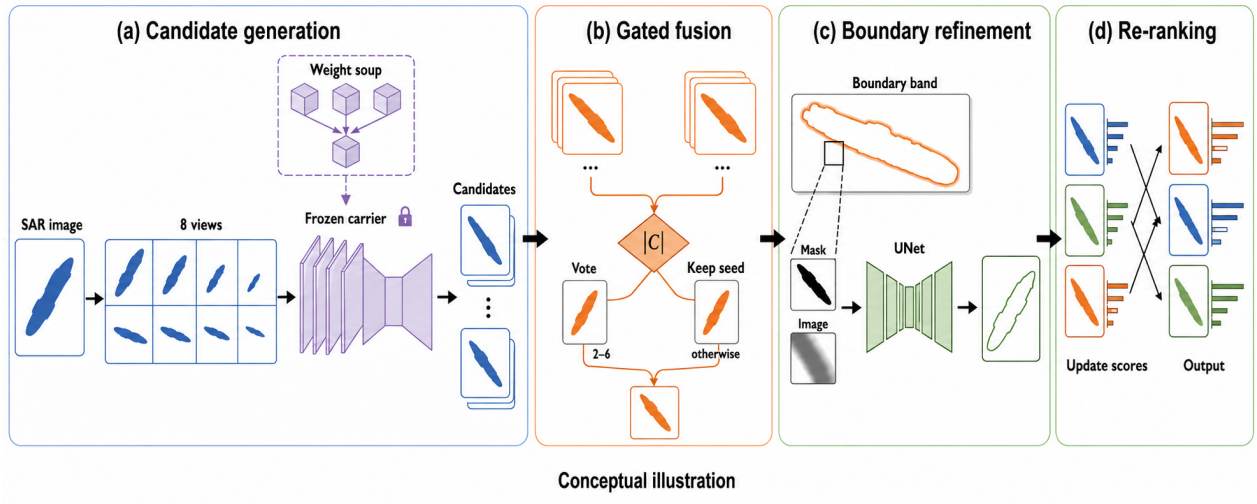

Fig. 1. Conceptual architecture of candidate generation, gated fusion, boundary refinement, and quality re-ranking. Checkpoints are averaged after SAR fine-tuning and remain frozen during augmented inference. HRSID training includes auxiliary mask-quality supervision; PSeg-SSDD uses the standard head. Fusion edits the candidate set, BPR edits covered boundary pixels, and re-ranking edits confidence. Tensor stacks, mask glyphs, and score bars illustrate computational roles and are not recorded observations or quantitative results. Internal network stages are abstracted; Supplementary Fig. S3 details the carrier readout.

[10]. It reads the SAR patch and coarse mask, and writes its prediction only inside a prescribed morphological band around the coarse boundary. A separate mask-quality regressor uses candidate-support, score, and geometry features to recalibrate the output scores. Both are fitted using predictions and existing annotations from the training split; neither updates the fitted carrier during postprocessing. The refiner preserves pixels outside its allowed band and leaves scores unchanged, whereas the regressor leaves masks unchanged. These interfaces make changes traceable, but do not by themselves imply additive gains or interchangeable execution order.

The experiments evaluate these choices through fusion controls on the same candidate pool, a progressive cascade, and repetitions with additional datasets and carriers. On HRSID, the reported official mask AP changes from 45.6 for the base carrier to 59.4 for the complete cascade, and inshore mask AP from 27.7 to 44.5. These are exploratory results: HRSID test subsets were used to select inference configurations. A later exact eight-view replay shows that a simpler per-view top-1 vote reaches 56.4 official AP before either corrector, compared with 53.8 for the selected gate. Applying the historically selected BPR weights to both outputs gives 58.81 versus 58.56 official AP; a refitted, out-of-fold top-1 cascade including re-ranking has not been assessed. On PSeg-SSDD, selected processing settings are reused while a standard-head carrier is newly trained and both correctors are fitted to that dataset; the reported official mask AP changes from 60.9 to 67.5. This tests the recipe under dataset-specific fitting, not frozen-model cross-dataset transfer or independent confirmation of every HRSID design conclusion. Literature-reported results are included as context; differences in initialization, model selection, and computational budgets preclude treating them as a single matched ranking.

The contributions are as follows.

- A documented comparison of mask-IoU association and candidate-cardinality-gated voting on augmented SAR predictions. The matched per-view top-1 control outperforms the selected gate on all three HRSID test partitions, so these observations do not support the gate as a necessary algorithmic contribution.
- A defined interface between fusion, an adopted boundary-patch refiner, and learned score calibration. Training-split predictions provide the correction inputs, existing labels provide supervision, and explicit write constraints delimit which parts of each output can change. The contribution is the specified integration and its evaluation, rather than the invention of mask voting or boundary-patch refinement.
- An empirical characterization of the cascade on HRSID and PSeg-SSDD, together with component and carrier controls. The study separates exploratory configuration selection, transferred settings, and literature context, and reports the observed scope of the gains without attributing them to a general preservation principle or an isolated pretraining effect.

Section II reviews related work. Section III defines the carrier, fusion rule, and correctors. Section IV describes the experimental protocols and results. Section V discusses interpretation and limitations, and Section VI concludes the article.

II. Related Work

A. SAR Ship Instance Segmentation

SAR ship instance segmentation combines recognition with spatial delineation under scattering-dependent appearance and coastal clutter. HRSID [1] and the pixel-level SSDD annotations [2] support evaluation beyond bounding-box detection. General instance segmenters, including Mask R-CNN [11] and Mask2Former [8], provide reusable foundations. SAR-specific approaches add mechanisms aimed at the imaging conditions or target geometry: LFG-Net uses low-level feature guidance and super-resolution denoising [3]; O2Former introduces query generation and orientation-aware modeling [4];

DiffSARShipInst uses diffusion-based instance decoding and asymmetric-convolution refinement [5]; and Gaussian-heatmap encoding offers an alternative to bounding-box-based instance representation [6]. SARSAM studies parameter-efficient adaptation of a foundation segmentation model [7].

Architectural design and training protocol both affect such comparisons. Work on SARDet-100K also studies pretraining transfer for SAR detection [12], emphasizing the relevance of the initialization regime. Our HRSID experiments use a COCO-initialized Mask2Former-R50 carrier trained with an auxiliary mask-quality objective, whereas the PSeg-SSDD carrier uses the standard head. Comparisons with results obtained using different initializations or schedules cannot isolate either pretraining or architecture as the cause of an AP difference. The focus here is the behavior of candidate aggregation and subsequent correction after this carrier has been fitted. It complements representation-level methods and does not establish that a stronger representation, different loss, or longer training schedule is generally unnecessary.

B. Candidate Aggregation and Mask Voting

Weight averaging and TTA provide established ways to obtain or combine predictions. Model soups average parameters of models fine-tuned from a common initialization [9]. TTA instead evaluates transformed inputs, requiring prediction alignment and an aggregation rule. In box space, Soft-NMS reduces scores according to overlap [13], whereas weighted boxes fusion combines box coordinates [14]. Learned weighting of augmented predictions has also been studied for classification [15]. These methods motivate careful treatment of aggregation as part of inference; they do not make an arbitrary merge rule appropriate for every candidate representation.

Mask voting has direct precedents in instance segmentation. MNC aggregates related mask predictions by classification-score-weighted pixel averaging after box-based suppression, then binarizes the result [16]. FCIS also forms final masks by weighted pixel-level aggregation of overlapping instance predictions [17]. Consequently, combining mask hypotheses is established practice. Relative to these predecessors, the present rule uses binary mask IoU for score-seeded association of aligned TTA candidates and conditions unweighted, foreground-tie-preserving voting on a finite cluster-size band. A singleton or a cluster above the band retains its seed. The proposed object of study is this particular association-and-selection rule in the evaluated SAR setting. Soup, TTA, pixel voting, and the broader idea of inference aggregation are adopted ingredients. Because cluster membership is not restricted to one candidate per view, the cardinality gate is also distinct from a gate defined on the number of agreeing views.

C. Boundary Correction and Mask-Quality Calibration

Boundary refinement addresses errors that may remain after an instance has been detected. PointRend predicts labels at selected points through a learned segmentation head [18]; SegFix estimates boundary-to-interior offsets for correcting segmentation outputs [19]. BPR extracts patches along predicted instance boundaries and applies a separate high-resolution refinement network to the output of an existing segmenter [10]. Our refiner adopts this BPR principle and the use of a coarse prediction as an input. Its specification for this study consists of SAR-intensity and coarse-mask channels, training-split prediction/label pairs, and a morphological restriction on where the patch predictions may be written back. Local re-segmentation and post-hoc boundary correction are therefore not claimed as new concepts. The band defines a preservation constraint on output pixels; whether it contributes to accuracy is an empirical question.

Spatial accuracy and confidence calibration are related but different objectives. IoU-Net estimates localization quality for boxes [20], and Mask Scoring R-CNN regresses mask IoU to improve alignment between mask quality and detection score [21]. The auxiliary training objective of our HRSID carrier belongs to this family of mask-quality supervision. The subsequent external re-ranker is a separate component: it estimates quality from saved support statistics, detection scores, and mask geometry, then blends that estimate with the existing score. It changes ordering without changing mask pixels. The refiner and re-ranker share a supervised correction workflow based on training annotations, while targeting different output variables. Their combined effect must be evaluated because changing a mask can alter its overlap with ground truth and changing scores can alter matching order. The interface provides traceable edits, rather than a proof that the two operations commute or improve every AP endpoint.

III. Method

A. Carrier and Inference Pipeline

Given a grayscale SAR chip $\mathbf{x} \in \mathbb{R}^{H\times W}$, replicated to three input channels, the pipeline returns scored instance masks $\mathcal{Y} = \{(\mathbf{m}_i, s_i)\}_{i=1}^{N}$, with $\mathbf{m}_i \in \{0,1\}^{H\times W}$ and $N \leq 100$. We separate offline model fitting from inference (Fig. 1). A trained carrier supplies candidates; mask-space fusion merges them; a boundary-patch refiner edits their masks; and a quality regressor adjusts their scores. The latter two models use existing training annotations to supervise errors in carrier predictions.

The carriers use the Mask2Former-R50 segmentation architecture [8], initialized from COCO instance-segmentation weights. For HRSID, the training objective is $\mathcal{L}_{\mathrm{M2F}} + \beta\mathcal{L}_{\mathrm{quality}}$, with $\beta = 0.5$. The auxiliary quality head reads the final query embedding and mask statistics and predicts the IoU of a matched mask with its training annotation, using a binary-cross-entropy objective. Its inputs are not detached, so this loss also updates shared representations. In the final inference recipe, auxiliary-head score mixing is disabled ($\lambda = 0$); this does not remove the auxiliary supervision used during training. The PSeg-SSDD carrier instead uses the standard Mask2Former head without this auxiliary objective. Both carriers are frozen after fitting, and neither postprocessor updates their parameters. Thus, "training-free" below refers to candidate generation and fusion from already trained checkpoints, not to the complete system. Training and selection protocols are specified in Section IV-C.

B. Cluster-Support-Gated Mask Fusion

1) Candidate generation: We adopt weight averaging [9] and multi-scale flip TTA to generate complementary predictions. Three checkpoints trained from the same initialization and parameterization are averaged offline,

$$\theta^\star = \frac{1}{3} \sum_{s \in \{s_0, s_2, s_3\}} \theta_s. \tag{1}$$

Here, θ_s denotes a member's parameter vector. Floating-point parameters and buffers are averaged; equal integer buffers are copied. The resulting checkpoint has the inference model count of one member, although obtaining the members incurs three training runs. Structural compatibility does not ensure beneficial weight averaging; Section IV-D1 reports the member-count study.

For HRSID, the eight views are

$$\mathcal{V} = \{640, 800, 960, 1088\} \times \{\mathrm{id}, \mathrm{hflip}\}, \quad V = 8. \tag{2}$$

Each view resizes the input and optionally flips it horizontally. Predicted masks are resampled to the native chip grid and unflipped before pooling. With at most 100 candidates per view, $K \leq 800$. PSeg-SSDD uses $\{416, 512, 608, 704\}$ with the same flip choices. A candidate retains its binary mask $\mathbf{m}_k$, score s_k, class c_k, and originating view v_k.

2) Mask-space clustering and readout: Nearby ships can have strongly overlapping boxes while occupying distinct pixels. We therefore associate candidates in mask space. Flattening the masks into the rows of $\mathbf{M} \in \{0,1\}^{K \times HW}$ gives

$$J_{jk} = \frac{\mathbf{m}_j^\top \mathbf{m}_k}{\mathbf{1}^\top \mathbf{m}_j + \mathbf{1}^\top \mathbf{m}_k - \mathbf{m}_j^\top \mathbf{m}_k}, \tag{3}$$

with $J_{jk} = 0$ for zero union. Intersections are computed by $\mathbf{M}\mathbf{M}^\top$. For the unused candidate indices $\mathcal{U}$, the highest-scoring seed defines

$$\begin{aligned} r &= \arg\max_{k \in \mathcal{U}} s_k, \\ \mathcal{C} &= \{r\} \cup \{j \in \mathcal{U} : c_j = c_r,\ J_{rj} \geq \tau\}, \end{aligned} \tag{4}$$

where $\tau = 0.5$. Members are then removed from $\mathcal{U}$. Association is greedy and seed-relative: members need not satisfy the threshold with one another. Multiple members may originate from one view. Consequently, the support $|\mathcal{C}|$ is candidate cardinality, distinct from the view count $n_v = |\{v_j : j \in \mathcal{C}\}|$.

For each cluster, a cardinality gate chooses pixel voting or seed retention:

$$\mathbf{m}^{\mathrm{out}}(p) = \begin{cases} \mathbf{1}\Big[\sum_{j \in \mathcal{C}} \mathbf{m}_j(p) \geq |\mathcal{C}|/2\Big], & 2 \leq |\mathcal{C}| \leq S_{\max}, \\ \mathbf{m}_r(p), & \text{otherwise}, \end{cases} \tag{5}$$

with $S_{\max} = 6$. A foreground tie is retained, so a two-member vote is the union. Singletons and clusters above the ceiling retain their seed. This rule limits averaging to a selected support range; support is a heuristic decision variable, not an estimate of statistical variance or a guarantee of mask quality. The deployed score is

$$s^{\mathrm{out}} = s_r. \tag{6}$$

Algorithm 1 Cluster-Support-Gated Mask Fusion

Require: native-grid candidates $\{(\mathbf{m}_k, s_k, c_k, v_k)\}_{k=1}^{K}$; $\tau = 0.5$; $S_{\max} = 6$
Ensure: scored fused masks $\mathcal{Y}$

```
compute J by Eq. (3); U ← {1, ..., K}; Y ← ∅
while U ≠ ∅ do
    select seed r and cluster C by Eq. (4)
    U ← U \ C
    read out m^out by Eq. (5); inherit s_r
    recompute box; retain cluster statistics and view count
    append (m^out, s_r) to Y
end while
return top-100 of Y by score
```

The implemented generalized rule $s_r(|\mathcal{C}|/V)^\gamma$ reduces to Eq. (6) at the deployed $\gamma = 0$; its ratio need not lie in $[0, 1]$. Boxes are recomputed from output masks, and the 100 highest-scoring clusters are retained. Algorithm 1 summarizes the procedure. The dense IoU computation has $O(K^2 HW)$ arithmetic cost; overall runtime also depends on the view resolutions and subsequent refinement workload.

Figure 2 visualizes the two readout branches. Supplementary Fig. S1 provides a restricted replay from stored masks; truncating that pool changes cluster support and may change the selected branch. Quantitative comparisons in Section IV-D2 use the complete candidate pool and the reported selection protocol.

C. Boundary-Patch Refinement

Fusion addresses candidate association but does not explicitly learn high-resolution boundaries. We adopt the boundary-patch refinement framework [10] with a local write constraint. The frozen souped carrier first predicts the training images in a single view. Predictions with score at least 0.2 are matched to training masks at IoU at least 0.5. Along matched contours, 64×64 image, prediction, and annotation patches are sampled with one center per 32×32 grid cell. Boundary pixels are scanned in row-major order; the first pixel in each grid cell is retained, with at most 32 patches harvested per instance. The resulting supervision uses existing annotations; it is not self-supervision or additional manual labeling.

A 2.35M-parameter UNet reads normalized grayscale and coarse-mask channels. Its four-down/four-up structure uses encoder widths 32/64/128/192/192, double convolution–batch normalization–ReLU blocks, and bilinear decoder upsampling with skip concatenation. Each patch is enlarged to 128×128, using bicubic image interpolation and bilinear mask interpolation, and the network is trained for 20 epochs with binary-cross-entropy and Dice losses. The HRSID harvest contains 37 009 patches from 10 297 matched instances on 3641 training images; 1850 patches form the checkpoint-selection holdout. This is a random 5% patch split, rather than an image-disjoint validation set. Remaining fitting details and selection limitations are given in Section IV-B and Section IV-C.

At deployment, instances with score below 0.05 pass through unchanged. For the remaining fused masks, the same deterministic sampling rule extracts at most 64 patches per

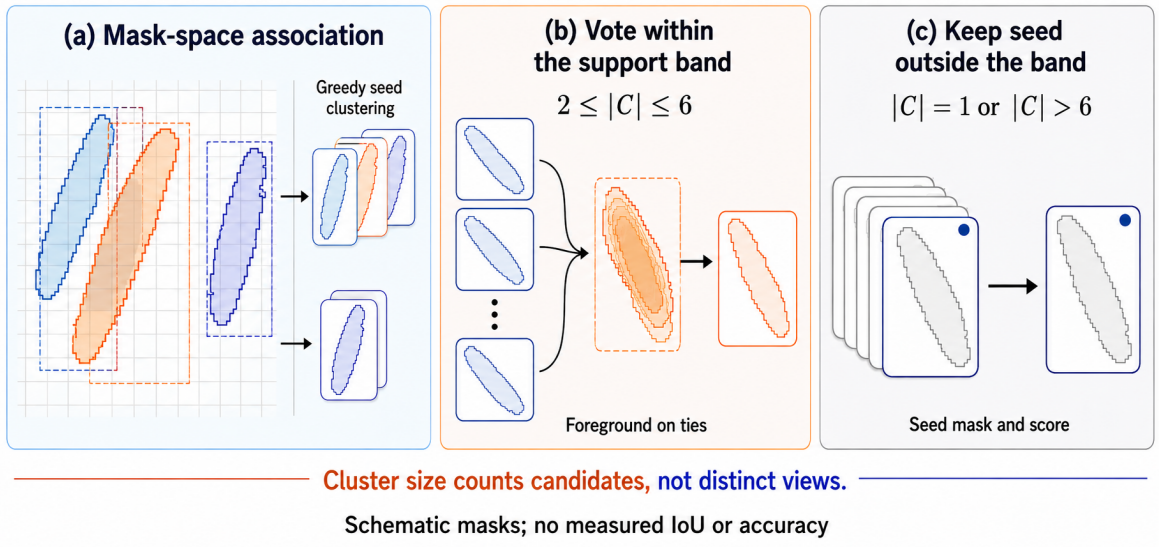

Fig. 2. Conceptual mask-space fusion. Score-ordered seed clustering uses mask IoU, then candidate cardinality selects pixel voting for $2 \le |\mathcal{C}| \le 6$ or seed retention otherwise. A foreground tie is retained and each output inherits its seed score. Candidate count differs from distinct-view count. Mask glyphs illustrate the operator and do not represent measured candidates or improvement against ground truth; a stored-candidate replay is provided in Supplementary Fig. S1.

instance; Eq. (7) applies to these processed instances. Refiner logits are passed through a sigmoid, area-resampled to 64×64, and averaged where patches overlap. Let $\bar{q}(p)$ denote this averaged probability and Ω the pixels covered by at least one patch. The writable band is the discrete morphological set $B_r = \mathrm{dilate}_r(\mathbf{m}^{\mathrm{out}}) \oplus \mathrm{erode}_r(\mathbf{m}^{\mathrm{out}})$, where $\oplus$ is exclusive-or and both operations use $r = 4$ iterations of a 3×3 square kernel. The output is

$$\mathbf{m}^{\mathrm{final}}(p) = \begin{cases} \mathbf{1}[\bar{q}(p) \ge 0.5], & p \in B_4 \cap \Omega, \\ \mathbf{m}^{\mathrm{out}}(p), & \text{otherwise.} \end{cases} \tag{7}$$

An empty refined result falls back to the input mask. This operation preserves scores, detection count and identities, and all pixels outside the writable covered band; it does not guarantee topology preservation or improved IoU for every instance. Single-view training and post-fusion deployment use different coarse-mask distributions, whose empirical transfer is evaluated in Section IV-D3.

D. Learned Mask-Quality Re-Ranking

Mask-quality scoring complements classification confidence [21]. Here, a separate regressor estimates quality from the fused prediction and its cluster statistics, without changing the carrier. For every stored pre-refinement fused candidate, we form

$$\begin{aligned} \boldsymbol{\phi} &= [\boldsymbol{\phi}_{\mathrm{c}}^{\top}, s^{\mathrm{out}}, \boldsymbol{\phi}_{\mathrm{g}}^{\top}]^{\top} \in \mathbb{R}^{14}, \\ \boldsymbol{\phi}_{\mathrm{c}} &= [|\mathcal{C}|, n_v, \bar{J}, J_{\min}, s_r, \Delta s]^{\top}, \\ \boldsymbol{\phi}_{\mathrm{g}} &= [\log(1+a), \log(1+w), \log(1+h), \\ &\qquad w/h, a/(wh), c_x, c_y]^{\top}. \end{aligned} \tag{8}$$

Here, $\bar{J}$ and $J_{\min}$ exclude diagonal self-IoUs and are zero for singletons; Δs is the member-score range. Geometry comprises mask area a, box width and height w, h, and normalized box center c_x, c_y, measured before BPR. Small denominators use numerical regularization. The repeated score entries agree under Eq. (6). No image intensity is read by this feature extractor.

Candidates with score at least 0.01 contribute training rows, including unmatched candidates. The target is maximum mask IoU with a training annotation, or zero if none overlaps. A histogram-based gradient-boosted regressor g is fitted on these rows. Its blend coefficient $\alpha \in \{0.3, 0.5, 1.0\}$ is selected using an image-disjoint 4:1 partition within the carrier's training split. Selection maximizes COCO mask AP on the held-out pre-BPR fused masks; original scores are evaluated as a reference but $\alpha = 0$ is not included in the selection loop. The regressor g is then refitted on all rows. The target measures individual mask overlap rather than the evaluator's one-to-one assignment of detections to annotations.

At application, BPR masks are associated with the stored pre-refinement candidates by score-ordered, greedy one-to-one matching at mask IoU at least 0.5. For a BPR record b, let $c(b)$ denote its matched pre-refinement candidate. This candidate supplies the original cluster and geometry features, and the record's score becomes

$$s_b^{\mathrm{final}} = \alpha\ \mathrm{clip}(g(\boldsymbol{\phi}_{c(b)}), 0, 1) + (1-\alpha) s_b^{\mathrm{in}}. \tag{9}$$

Here, s_b^{in} is the score already attached to the BPR record, which BPR leaves unchanged; its feature vector comes from the matched pre-refinement candidate. The selected blend is $\alpha = 0.5$ for HRSID and $\alpha = 1.0$ for PSeg-SSDD. Unmatched records retain their scores. All masks, detection count, and image/class identities are preserved; only scores change. The re-ranker therefore changes only scores, whereas BPR changes only masks; nevertheless, refined geometry affects candidate association, and AP depends jointly on masks and ranking. We evaluate their sequential combination without assuming commutativity or guaranteed gains. Each deployed carrier–dataset pair requires its own fitted correctors; the full three-

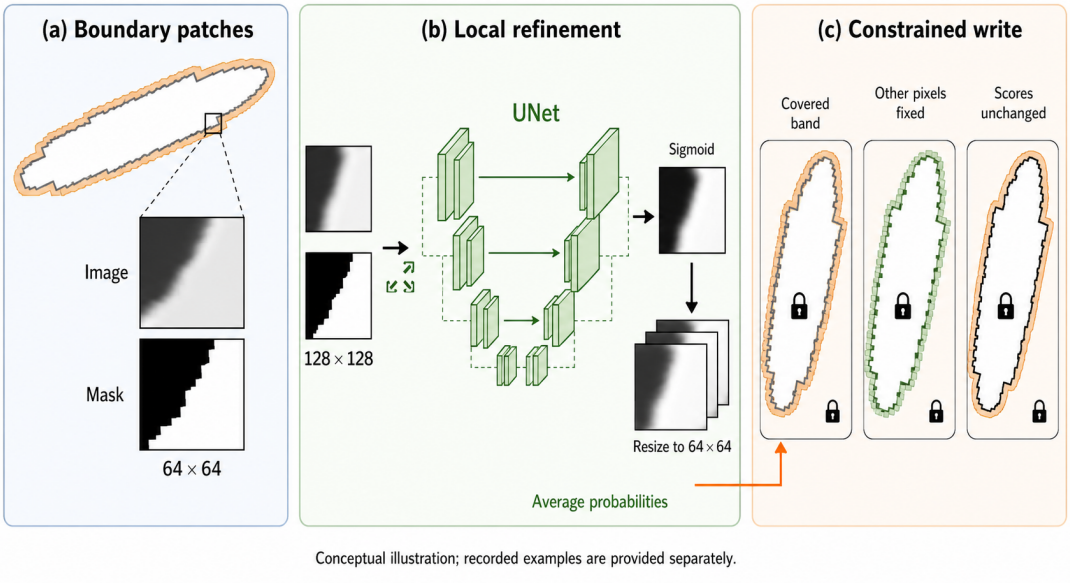

Fig. 3. Conceptual boundary-patch refinement. Native 64×64 grayscale and coarse-mask patches are resized to 128×128 for the UNet. Sigmoid probabilities return to the native grid and are averaged across overlapping patches. Only covered pixels in the discrete morphological band may change; other pixels, scores, and detection records are retained. The band uses four iterations of square-kernel dilation and erosion. This schematic is complemented by a recorded single-view example in Supplementary Fig. S2.

stage cascade and the subset transferred to other carriers are distinguished in Section IV-F.

IV. Experiments

We first state the data, carrier training, and selection procedures, then examine changes within a fixed carrier and candidate pool. Literature-reported results and local reimplementations provide context under different training budgets. HRSID fusion settings were selected on its test data; those results are therefore exploratory rather than independent confirmation.

A. Datasets

HRSID [1] contains 5604 annotated 800×800 SAR chips, with 3642 training and 1962 test images. The test partition used by prior work [22] contains 369 inshore and 1593 offshore chips. PSeg-SSDD contains 928 training and 232 test images at 512×512, including 46 inshore and 186 offshore test images. "Official" denotes the full test set; its inshore and offshore subsets are not independent replications. In our local audit of the complete HRSID annotation files, 136 of 137 filename parent-image prefixes occur in both official partitions, and 1820 of 1962 test chips geometrically overlap at least one training chip with the same prefix. These prefixes identify numbered parent images, not verified satellite acquisition products; geometric overlap does not by itself measure its effect on AP. A separate HRSID ship-detection study likewise notes that its chip-disjoint partition is not source-scene-disjoint [23]; that study uses a reduced fixed subset rather than the complete chip counts audited here. The present evaluation is therefore chip-level and does not establish parent-image- or acquisition-disjoint generalization.

B. Carrier and Corrector Training

The HRSID c0 carrier is COCO-initialized Mask2Former-R50 [8] with 100 queries, trained with an auxiliary mask-quality loss. The saved HRSID configurations use mask-IoU loss weight 0.5. Setting the inference blend to $\lambda = 0$ disables auxiliary rescoring, not its earlier contribution to training. Thus, the carrier is frozen during the evaluated postprocessing, but is not a stock model trained without auxiliary supervision. Table VIII explicitly identifies a separate $\lambda = 0.2$ reference row. In contrast, the saved PSeg-SSDD member configurations use the standard Mask2FormerHead without this auxiliary objective. The datasets therefore share a model family and COCO initialization, not an identical training head or loss.

HRSID training uses AdamW, initial learning rate 2.5×10^{-5} with step decay, batch size 2 and 2-step gradient accumulation (effective batch 4), for 100 872 iterations or 55 epochs. PSeg-SSDD uses batch 4 without accumulation for 8352 iterations or 36 epochs. Computation uses a single NVIDIA RTX 3090 Ti. The main soups average seeds s0/s2/s3, requiring 3×55 and 3×36 carrier epochs, respectively. The HRSID view set is $\{640, 800, 960, 1088\} \times \{\text{identity}, \text{flip}\}$; PSeg-SSDD uses $\{416, 512, 608, 704\}$ with the same support gate $2 \le |\mathcal{C}| \le 6$.

BPR uses native 64-pixel boundary windows resized $2\times$ and 20 training epochs, reported as approximately 19 minutes on HRSID. The refiner uses AdamW with batch size 64, learning rate 10^{-3}, weight decay 10^{-4}, and cosine decay to 10^{-5} over 20 epochs. Training patches receive independent horizontal and vertical flips with probability 0.5 and a random multiple-of-90-degree rotation. The 5% patch holdout uses a fixed permutation seed of 123; the checkpoint with minimum holdout BCE-plus-Dice loss is retained. Its random patch-level holdout can contain overlapping windows from the same training image as the fitting partition; it is not image-

independent validation. The re-ranker uses histogram gradient boosting with 300 iterations, learning rate 0.08, and least-squares loss. It is fitted to 51 781 HRSID candidate rows from the 3642 training images, or 5122 rows from 928 PSeg-SSDD images, in the reported 19 s and 9 s on CPU. The blend $\alpha \in \{0.3, 0.5, 1.0\}$ is selected using a 4:1 image-identifier partition of these training data, giving 0.5 and 1.0, before refitting on all rows. Held-out Spearman correlations between predicted quality and target mask IoU are 0.68 and 0.86, respectively. Although the re-ranker split separates images, the carrier has already seen them; these are in-sample carrier residuals, not out-of-fold predictions.

Eight views specify an inference configuration, not an eightfold measured latency. Fusion forms a Gram product over $K \leq 800$ masks; BPR adds a variable number of patch evaluations and re-ranking adds a tree evaluation per instance. Synchronized stage timings, transfer overhead, and memory use are needed for a deployment-cost comparison. The Cascade R-CNN experiments also require a parameter-free repair of the framework's flipped-mask merge routine, reported as output-identical to the stock routine on non-flipped inputs.

C. Selection and Statistical Scope

Unless otherwise specified, results are COCO mask AP $\times 100$, rounded to one decimal, in inshore/official/offshore order. The three HRSID carrier seeds yield 27.7±0.0/45.7±0.2/59.8±0.3 (mean±sample s.d.); Table I uses seed s0, 27.7/45.6/59.6. This observed variation does not define a universal significance threshold. Deterministic inference reproduces BPR results to four decimals and re-ranking outputs byte-identically, but it does not eliminate training or selection uncertainty.

The main carrier checkpoints use the final iteration. The test-logging hook did not select those checkpoints; test data nevertheless influenced subsequent development. Specifically, $\tau = 0.5$, $S_{\max} = 6$, $\gamma = 0$, the view scales, and the fusion operator were chosen through HRSID test-set sweeps. The narrow ceiling sweep, 38.4/38.6/38.8 inshore with offshore 64.8, does not bound optimism from the full set of choices. A later probe on 728 previously seen training chips (20%) yielded 57.6 for all $S_{\max} \in \{4, 5, 6, 7\}$ and 57.7/57.6/57.4 for $\tau = 0.4/0.5/0.6$; this illustrates a different residual distribution and cannot validate test-set selection. Fusion settings were subsequently reused on PSeg-SSDD and other carriers. Correctors were refitted, and the reported BPR seed s0 was the best of three inspected runs. Competitor and long-schedule checkpoint-selection differences are stated with their results.

Quoted intervals use 1000 paired image-bootstrap resamples, with the 2.5th and 97.5th percentiles of each AP difference. They condition on the fixed predictions and selected settings. They do not cover hyperparameter selection, carrier/corrector training variation, or correlation among chips from a parent scene. Independent confirmation requires locked settings and image- or scene-disjoint development and evaluation data.

D. Within-Carrier Ablations

1) Candidate Generation and Soup Compatibility: Relative to seed s0, the gated stack in Table I improves AP by 11.1/8.2/5.2. Six-view TTA gains across the three individual seeds are 6.9/7.4/6.7 (mean 7.0, s.d. 0.3). The soup and six-view results are approximately additive in this comparison: the separate inshore increments 1.9 and 6.9 predict 36.5, versus the observed 36.1. A three-model ensemble over the same eight views gives 39.3/53.6/64.0 versus 38.8/53.8/64.8 for the soup. The soup evaluates one averaged model per view instead of three [9].

Table II shows that additional soup members can reduce AP. PSeg-SSDD soups with $N = 2$ and $N = 5$ fall below all their members on all splits. The $N = 3/4$ soups give offshore 64.8/65.5, above the member maximum 64.3, and official 60.9/61.6, above the six-seed mean 60.7, but inshore 51.9/52.3 remains below 52.9±0.7. The three main members were fixed before this sweep. Pairwise analysis separates $\{s_0, s_3, s_4, s_6\}$ and $\{s_2, s_5\}$: seven within-group pairs improve over their member mean by 0.3–0.8/0.8–1.4/0.8–1.8, whereas all eight cross-group pairs lose on inshore and official, including 5.0–12.7 inshore points. Parameter distances of 10.0–10.8, relative to $\|\theta_{s_0}\| = 300.3$, and fine-tuning-increment cosines of 0.46–0.52 do not separate these groups. Matching initialization and tensor structure is therefore insufficient to certify a useful soup. The HRSID three-member soup improves inshore AP by 1.9; a development-only compatibility rule remains to be established.

2) Fusion on a Fixed Candidate Pool: Table III changes only the merge rule on the same soup/eight-view candidates. Mask-space seed selection improves inshore AP from the box-NMS value 37.1 to 38.2; the support gate adds a further 0.6 to reach 38.8. Thus, the complete 1.7/1.2/0.6 advantage over box-NMS includes both mask-space clustering/selection and conditional voting. An exact replay of the original 1962-image forward pass reproduced all 118 922 historical gated detections. Selecting the highest-scoring candidate from each represented view within those same seed clusters and applying a fixed vote changes only their masks, preserving the number, score, category, and order of outputs. This stronger simple control reaches 41.0/56.4/67.8, exceeding the gate by 2.2/2.6/3.0 mask AP points. This diagnostic uses the historically inspected test set and is not independent confirmation.

Applying the historically selected BPR checkpoint to the same top-1 output gives official mask AP 58.8101, versus 58.5646 for gated fusion followed by that refiner. Inshore AP is 44.2000 versus 43.5540; offshore AP is 69.0596 versus 69.1425. Candidate identity, class, score, and order remain fixed through this comparison; only masks differ. The 2.5834-point official advantage before BPR thus shrinks to 0.2455 after BPR, and the offshore ordering reverses. These figures reuse a refiner trained on in-sample carrier predictions and selected after historical test inspection; they neither isolate a causal gate benefit nor compare a newly refitted top-1 BPR and re-ranking cascade.

Among the originally tested rules, unconditional pixel voting gives the highest inshore entry, 39.1, but reduces offshore

TABLE I
PROGRESSIVE POSTPROCESSING RESULTS ON HRSID (MASK AP $\times 100$). THE CARRIER IS FROZEN AFTER AUXILIARY-SUPERVISED TRAINING. SOUP USES THREE TRAINED MEMBERS; THE CORRECTORS ARE FITTED SEPARATELY

Configuration	Additional fitting	Views / postprocessing	Inshore			Official (full)			Offshore		
			AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
c0 base, COCO-init M2F-R50 (seed s_0)	55 ep	1 view	27.7	61.8	22.1	45.6	81.9	49.4	59.6	97.2	71.2
+ weight soup (avg. of s0/s2/s3)	2 further seeds	1 view	29.6	–	–	47.4	–	–	61.1	–	–
+ 6-view TTA $\{736, 800, 864\}\times$flip	0	6 views	36.1	–	–	52.5	–	–	64.4	–	–
+ wide 8-view TTA, box-NMS	0	8 views	37.1	–	–	52.6	–	–	64.2	–	–
+ cluster-support-gated fusion	0	8 views	38.8	75.6	36.1	53.8	89.4	62.2	64.8	98.5	81.7
+ BPR refiner	19 min	8 + BPR	43.6	76.3	47.0	58.6	89.9	70.7	69.1	**98.6**	87.4
+ learned re-ranking	19 s	8 + BPR + rank	**44.5**	**76.7**	**48.3**	**59.4**	**90.1**	**71.4**	**70.1**	**98.6**	**88.0**

Bold denotes the best value in each column; ties at one decimal are both marked. "–": complete metric rows are shown for the single-view base and three cascade endpoints. Each carrier member is trained for 55 epochs; the soup therefore requires 3×55 epochs in total. Zero in the additional-fitting column denotes no fitting for that operation, not zero total training cost. The 19-min and 19-s entries are reported corrector-fitting times. View counts do not substitute for measured end-to-end latency. The last three rows use the same already trained carrier.

TABLE II
EXPLORATORY WEIGHT-SOUP MEMBER-COUNT SWEEP N ON PSEG-SSDD (MASK AP $\times 100$, SINGLE-VIEW BASE, NO TTA). MEMBERS ARE ADDED CUMULATIVELY IN THE FIXED SEED ORDER $s_0, s_2, s_3, s_4, s_5, s_6$

N	Members	Inshore	Official	Offshore
1	s_0	53.1	60.2	63.3
2	s_0, s_2	48.3	57.2	61.6
3	s_0, s_2, s_3	51.9	60.9	64.8
4	s_0, s_2, s_3, s_4	52.3	61.6	65.5
5	s_0, s_2, s_3, s_4, s_5	45.2	55.9	61.4
6	$s_0, s_2, s_3, s_4, s_5, s_6$	48.0	59.4	64.2
Six single seeds, mean $\pm$ s.d.		52.9 $\pm$ 0.7	60.7 $\pm$ 0.3	64.1 $\pm$ 0.4

Each member is a separately trained 36-epoch carrier. The main three-member soup was fixed before this test-set sweep. These rows illustrate non-monotonic averaging behavior; they do not provide a test-independent rule for choosing N or member identities.

TABLE III
FUSION OPERATORS ON IDENTICAL SOUP $\times$ 8-VIEW CANDIDATES (MASK AP $\times 100$)

Fusion operator	Inshore	Official	Offshore
box-NMS (reference)	37.1	52.6	64.2
matrix-NMS (tested configuration)	4.1	8.9	14.4
unconditional pixel vote	39.1	48.7	55.9
probability mean	39.1	48.7	55.9
seed vote	38.2	53.5	64.8
support-gated vote ($S_{\max} = 6$)	38.8	53.8	64.8
per-view top-1 vote	**41.0**	**56.4**	**67.8**

Bold denotes the best value in each column; top-1 is a later exact replay. Exploratory test-set ceiling sweep: $S_{\max} = 4/5/6$ gives inshore 38.4/38.6/38.8, offshore pinned at 64.8 throughout.

AP to 55.9, below the 59.6 single-view baseline. The gated result 38.8/53.8/64.8 preserves the seed-only offshore result while increasing inshore AP_{75} from 35.3 to 36.1; both are below the later matched top-1 result. The tested probability-mean control has the same rounded metrics as unconditional voting; this does not establish bitwise or mathematical equivalence. Matrix-NMS gives 4.1/8.9/14.4 in the tested configuration. This is a failed setting requiring its kernel, score filter, and top-K behavior to be checked, not evidence that Matrix-NMS inherently retains all duplicates or is generally unsuitable.

With the same gated rule, two views ($\{800\}\times$flip) give 33.6/50.5/63.4, four ($\{800, 960\}\times$flip) give 38.3/53.8/64.9, and eight give 38.8/53.8/64.8. Four views are within 0.5 inshore AP of eight. This suggests four views as a useful operating point for a measured compute/accuracy study.

3) Boundary Correction and Score Re-Ranking: BPR adds the reported 4.8/4.7/4.4 AP, using unrounded differences of 4.76/4.72/4.39. The paired 95% intervals are $[4.05, 5.55]$, $[4.25, 5.22]$, and $[3.81, 4.89]$, respectively, positive in all 1000 fixed-prediction resamples. Inshore AP_{50} increases by 0.7, whereas AP_{75} increases by 10.9 (8.4 official and 5.7 offshore). AP_s changes from 35.8/52.8/64.1 to 41.4/57.8/68.7. The concentration at stricter overlap is consistent with improved boundaries.

The reported application checks find no annotation dependency in prediction generation, MD5-identical regeneration, zero out-of-band edits in 90/90 sampled instances, and zero changes among roughly 237 000 scores across the overlapping evaluation splits. Training/test filename overlap is zero. These checks establish specific dataflow and write-field properties, not complete protection from split, selection, or parent-scene leakage. Applying the same refiner with $r = 2/4/8$ or no band gives 43.6/58.6/69.1 in all cases, with differences below 0.01 AP. Thus, the band guarantees where writing is permitted; these results do not attribute the measured gain to that guard. The other training seeds give 43.1/58.1/68.7 and 43.2/58.3/68.8, yielding a three-seed mean 43.3±0.2/58.3±0.2/68.9±0.2. The reported s0 refiner is 0.3–0.5 AP above those runs, and the following re-ranking results inherit this selection.

Re-ranking adds 0.9/0.8/1.0 AP in the rounded presentation (0.97/0.82/0.95 before rounding), with paired intervals $[0.61, 1.23]$, $[0.61, 1.00]$, and $[0.63, 1.18]$. All 1000 paired differences are positive for the selected predictions. AP_s reaches 42.0/58.3/69.4, while large-object AP falls by 1.4/1.3/1.3. The latter is a failure mode requiring size-stratified calibration analysis; the predominantly small-vessel training distribution is a possible explanation. The fitted regressor uses training predictions/annotations and prediction-derived features, and its output leaves mask bytes unchanged.

4) Exploratory Architecture and Training Controls: Table IV retains the attempted controls without treating them

TABLE IV
EXPLORATORY ARCHITECTURE AND TRAINING CONTROLS (MASK AP ×100). UPPER BLOCK: VERSUS THE C0 BASE; LOWER BLOCK: VERSUS THE FULL METHOD

Configuration	Inshore	Official	Offshore
c0 base (3-seed reference)*	27.7	45.7	59.8
OffsetAdj (constrained offsets, 1 seed)	26.7	45.3	59.7
CHART (aux. supervision, w=0.1, 1 seed)	≈27.7	45.7	59.8
CAST (hard-sample paste augm., 1 seed)	27.7	–	–
Training-side families (4)	reported as flat		
Mask-IoU head (λ=0.2, seed s_0)	29.2	47.3	62.0
SGA (zero-gated strip adapter, 1 seed)	26.8	45.5	60.4
Swin-S backbone swap (1 seed)	28.6	46.1	59.9
Swin-S + recovery stack (1 seed)	37.3	52.5	63.5
200 ep + recovery stack (1 seed)	37.9	52.5	63.5
R-50 gated stack (3-seed soup)	**38.8**	**53.8**	**64.8**

Bold denotes the best value in each column. The seven directions are OffsetAdj, CHART, CAST, the four training-side families, the mask-IoU head, SGA, and the Swin-S swap; the 200-epoch row is a longer-training control with different scheduling and checkpoint selection. The lower-block controls use one seed, whereas the R50 reference uses a three-member soup. These rows do not isolate backbone capacity or training duration. *Official/offshore means of the 3-seed control axis, with observed seed variation up to 0.3 AP relative to the seed-s_0 figures 45.6/59.6 of Table I.

as a causal architecture study. OffsetAdj and the zero-gated SGA strip adapter [24], [25] reduce inshore AP by 1.0 and 0.9, respectively; SGA improves offshore AP by 0.6. CHART fails at auxiliary weight 1.0 and is flat at 0.1; CAST and four grouped training-side families are reported as flat in the available record. Auxiliary inference rescoring at $\lambda = 0.2$ adds 1.5 inshore AP alone and leaves a 0.1 residual under the stack. This concerns use of the already trained quality head and does not establish that auxiliary training was unnecessary.

Swin-S [26] has a reported 3.2-point COCO advantage and gives 0.9 inshore AP above the R50 carrier. Its single-seed stack scores below the three-member R50 stack, so that comparison confounds capacity and seed averaging. The 200-epoch run saturates near epoch 45 and reaches an official base best of 46.4, a 0.8 increase concentrated at AP_{75}, using an early-decayed warm restart and best-on-official-test selection. Its stack trails by 0.9/1.3/1.3. Different scheduling, checkpoint selection, and soup budgets preclude attributing these differences solely to duration or preservation of pretrained representations.

E. Comparative Results

1) Literature-Reported HRSID Results: Literature values retain the arXiv v1 source of O2Former [22]; that work was subsequently published in JSTARS [4]. Table V juxtaposes these reported values with our evaluated cascade; it is not a common-code rerun. The full cascade reports 44.5 inshore AP, 48.3 AP_{75}, and 42.0 AP_s; the corresponding listed O2Former values are numerically 1.4, 2.4, and 2.1 lower under a different protocol. Offshore, the cascade's 70.1 AP is below the reported 71.7 and 70.9 of O2Former and DiffSARShipInst; AP_{50} is 98.6. Before BPR, the stack has inshore AP_{50} 75.6 versus O2Former's 69.5, but AP/AP_{75} 38.8/36.1 versus 43.1/45.9. This pattern motivates examining boundary correction separately from candidate merging, without implying a matched comparative advantage.

These differences do not establish state-of-the-art performance under a matched budget. The 500-epoch ImageNet-backbone/from-scratch-head literature protocol differs from our COCO initialization, auxiliary supervision, three carrier trainings, and augmented inference. The reported Mask2Former baseline difference, 17.9 versus 27.7 inshore (9.8 points), is consequently not an isolated pretraining effect. The DiffSARShipInst entries, including offshore AP_{50}/AP_{75} of 97.8/90.0, are secondary-source values. No competitor official-split value is available in this comparison.

2) Local PSeg-SSDD Comparison: All rows of Table VI use the same training split and COCO evaluator, but initialization, epoch count, checkpoint selection, seed budget, and inference computation differ. Competitor checkpoints are best-on-test; the two SAR-module reimplementations were selected on inshore data. Their values are conditioned on test-based checkpoint selection. The modules are local implementations and should not be equated with the original complete systems.

The generic rows report 38.9–41.2 inshore AP, the local NDM and OAEM variants 52.9/53.2, and the full cascade 62.1/67.5/69.8. The corresponding margins over the strongest listed local competitor are 8.9/6.7/4.7. Applying the recipe to OAEM itself gives 54.3/60.6/63.3 after the stack and 59.9/64.2/66.1 after BPR, a 6.7-point inshore endpoint gain and AP_{75} change from 55.9 to 74.9. Its intermediate official/offshore changes are negative ($-0.2/-0.7$), demonstrating that the stack can reduce AP.

Table VII separates detection probability from false alarms. At score 0.30, BPR detects 169 of 172 inshore ships ($P_d = 0.983$), versus 159 ($P_d = 0.924$) for the NDM reimplementation, but produces more false alarms. The per-method optimum F_1 is 0.902 for BPR and 0.882 for OAEM; these thresholds are selected on the evaluated split and are descriptive, not deployment thresholds. At score 0.30 offshore, BPR, NDM, and OAEM each detect 369 of 374 ships. In Fig. 4, BPR reaches full recall with nonzero precision at IoU 0.50; AP_{50} is 95.7 versus a competitor maximum of 92.2, and AP_{75} is 72.5 versus 58.8. The curves stop before learned re-ranking.

3) Selected Qualitative Examples: Figures 5 and 6 use a favorable-case selection rule. For each inshore image, the stronger competitor minimizes missed targets plus half the false alarms; images are ranked by the cascade's reduction in that count, with false-alarm difference breaking ties. The two leading cases are shown. The offshore example maximizes inter-method mask-IoU spread among error-free cases. SOLOv2, with 19.0 HRSID inshore AP, is omitted from the wide qualitative comparison.

On the selected 95-ship HRSID chip, the stack detects 66 ships versus 23/28 for the local competitors; the two inshore examples show 1.9–2.9 times as many detections. Their shared score threshold does not equalize calibration across methods. The selected offshore mean matched IoUs are 0.782/0.777 for the competitors, 0.828 for the stack, and 0.873 after correction. In PSeg-SSDD, only one of 46 inshore images has a strictly positive count gain over the best competitor: the 13-ship example, detected at 13/13 by both displayed cascade variants. The second example is a count tie. The offshore mean matched IoU is 0.898 versus a competitor maximum of 0.888.

TABLE V
HRSID LITERATURE CONTEXT AND THE EXPLORATORY CASCADE (MASK AP $\times 100$). COMPETITOR ROWS AS REPORTED UNDER DIFFERENT SELECTION AND COMPUTE PROTOCOLS; "SCRATCH" DENOTES AN IMAGENET BACKBONE WITH A FROM-SCRATCH SEGMENTATION HEAD; "N/R" DENOTES NOT REPORTED

				Inshore				Offshore			
Method	Init	Epochs	Views	AP	AP_{50}	AP_{75}	AP_s	AP	AP_{50}	AP_{75}	AP_s
Gated stack + BPR + re-ranking (ours)	COCO	3×55	8	44.5	76.7	48.3	42.0	70.1	98.6	88.0	69.4
Gated stack + BPR (ours)	COCO	3×55	8	43.6	76.3	47.0	41.4	69.1	98.6	87.4	68.7
O2Former† [22]	scratch	500	1	43.1	69.5	45.9	39.9	71.7	97.5	91.0	71.1
DiffSARShipInst‡ [5]	n/r	n/r	1	42.6	76.1	44.1	40.3	70.9	97.8	90.0	70.3
Gaussian-Heatmap§ [6]	n/r	n/r	1	42.5	79.1	39.9	41.0	68.0	97.7	85.1	67.4
RTMDet† [27]	scratch	500	1	40.4	75.9	38.7	39.3	67.4	97.7	83.7	66.8
HQ-ISNet† [28]	scratch	500	1	39.6	72.5	41.0	39.0	67.9	97.5	87.1	66.5
Gated stack (ours)	COCO	3×55	8	38.8	75.6	36.1	35.8	64.8	98.5	81.7	64.1
Cascade Mask R-CNN† [29]	scratch	500	1	35.6	59.1	40.1	32.6	67.4	96.8	85.7	66.7
Mask R-CNN† [11]	scratch	500	1	34.0	56.5	38.3	31.0	65.3	95.8	83.4	64.5
c0 base, seed s_0 (ours)	COCO	55	1	27.7	61.8	22.1	24.3	59.6	97.2	71.2	58.6
Mask2Former† [8] (their baseline)	scratch	500	1	17.9	36.9	16.2	14.7	60.6	95.2	74.9	59.5

Values are descriptive context, not a matched method ranking. All methods use a ResNet-50 backbone where reported. Epochs: training budget, 3×55 denoting three 55-epoch seeds averaged into one weight soup. Views: number of inference views, not measured latency. Corrector fitting is additional to the stated carrier epochs. Competitors are listed as single-view where no TTA is reported; their cost is not directly comparable. †As tabulated in the arXiv v1 of O2Former [22], Table III; not re-evaluated by us. ‡Values relayed through O2Former's table; these detailed metrics have not been independently verified against the original system. §From [6], Table II; protocol not reported.

TABLE VI
LOCAL COMPARISON WITH SIX METHODS ON PSEG-SSDD (MASK AP $\times 100$). EVERY ROW, COMPETITORS INCLUDED, WAS TRAINED AND SCORED BY US UNDER ONE EVALUATOR. TRAINING AND CHECKPOINT SELECTION DIFFER ACROSS ROWS

Method	Family	Init	Epochs	Inshore	Official	Offshore
YOLACT [30]	single-stage prototype	ImageNet	55 (best@31)	38.9	55.7	63.4
Mask R-CNN [11]	two-stage region-based	ImageNet	50 (best@41)	40.8	57.8	65.1
Mask R-CNN + DCN [31]	two-stage + deformable conv. (DCN)	ImageNet	50 (best@19)	41.0	57.8	64.6
CondInst [32]	single-stage dynamic kernel	COCO	36 (best@36)	41.2	54.3	60.1
M2F + DenoDet-style NDM (our reimpl.)* [33]	query-based + SAR module	COCO	36 (best@27)	52.9	60.5	63.7
M2F + OAEM (our reimpl.)* [22]	query-based + SAR module	COCO	36 (best@36)	53.2	60.8	64.0
+ stack (1 seed)	+ training-free stack	COCO	36 (best@36)	54.3	60.6	63.3
+ stack + BPR (1 seed)	+ boundary refiner	COCO	36 (best@36) +20 BPR	59.9	64.2	66.1
Soup base (ours)	query-based M2F	COCO	3×36	51.9	60.9	64.8
Gated stack (ours)	+ training-free stack	COCO	3×36	54.0	63.3	66.9
Gated stack + BPR (ours)	+ boundary refiner	COCO	3×36 +20 BPR	60.0	66.3	68.8
Gated stack + BPR + re-ranking (ours)	+ learned re-ranking	COCO	3×36 +20 BPR	**62.1**	**67.5**	**69.8**

Bold denotes the best value in each column. All rows were fine-tuned on the identical 928-image training split and scored with the identical COCO evaluator at 512×512; the ImageNet-initialized group received the larger epoch budget. Checkpoints: all competitor rows use their best-on-test checkpoint, the epoch in parentheses, and the two reimplemented rows were selected on the inshore split they lead; our carrier rows use the final iteration. This is a difference in selection protocol, not a matched training comparison; initialization, training budget, soup membership, and test-time computation also differ. *These two rows use our implementations of the described modules on our Mask2Former carrier; they evaluate those local implementations, not the complete original systems. The two rows beneath the OAEM row apply our recipe to that same OAEM checkpoint: the training-free stack, then a refiner fitted on its own training-split predictions (8013 patches, 20 epochs); single seed, no soup. For reference, O2Former reports on this dataset, under its own 500-epoch from-scratch protocol, a Mask2Former baseline of 52.5 inshore and 67.2 offshore, and 55.2 and 71.0 with OAEM added [22]. The following attempted comparisons were excluded because of implementation or resolution problems: SOLO diverged, PointRend and InstaBoost were trained at 1333×800 but tested at 512×512 (a resolution artifact), and FreqFusion collapsed in its own training log.

F. Repetition Across Datasets and Carriers

1) PSeg-SSDD With Dataset-Specific Fitting: The HRSID fusion rule is reused with proportionally adjusted view sizes, but a new PSeg-SSDD standard-head carrier is trained and BPR and re-ranking are refitted on PSeg-SSDD training predictions. This is a repetition of the recipe under dataset-specific fitting, not frozen-model cross-dataset transfer. Table VI moves from the soup base 51.9/60.9/64.8 to 62.1/67.5/69.8, a combined 10.2/6.6/5.0 increase. The inshore stack increment is 2.1 versus 11.1 on HRSID; BPR contributes 6.0 of the 10.2 and moves AP_{75} from 54.9 to 72.5, followed by 74.6 with re-ranking. Official AP_{75} moves from 82.9 to 84.9 with re-ranking. The stage-dependent gains differ substantially between the two datasets.

Uncertainty is material on 46 inshore images: base-to-stack is 2.02 with interval $[-0.78, 4.36]$ (93% positive resamples), BPR adds 6.09 with $[2.06, 9.30]$, and re-ranking adds 2.09 with $[-0.09, 3.87]$ (97% positive). The first and third intervals include zero. On 186 offshore images the corresponding intervals are $[0.93, 3.23]$, $[0.74, 2.81]$, and $[0.21, 1.68]$. These conditional results do not establish significant improvement from every stage on every split. For context, O2Former reports a 52.5 inshore Mask2Former baseline and 63.2/71.9 for its complete model under a different 500-epoch protocol. The present cascade remains 1.1/2.1 below those complete-model values, while its soup is below the six-member single-seed mean 52.9 ± 0.7. The cross-paper HRSID baseline gap does not repeat here.

TABLE VII
FALSE-ALARM / MISSED-DETECTION ACCOUNTING ON PSEG-SSDD (MASK IOU ≥ 0.5, GREEDY SCORE-DESCENDING MATCHING) AT A FIXED SCORE THRESHOLD OF 0.30 AND AT THE PER-METHOD, PER-SPLIT TEST-SELECTED F_1-OPTIMAL THRESHOLD

	Threshold 0.30			F_1-optimal threshold			
Method	P_d	FA	F_1	P_d^*	FA*	F_1^*	thr*
Inshore split (46 images, 172 GT ships)							
YOLACT	0.709	0.957	0.722	0.692	0.587	0.748	0.40
Mask R-CNN	0.721	1.326	0.695	0.663	0.587	0.728	0.70
Mask R-CNN + DCN	0.738	1.739	0.670	0.663	0.717	0.715	0.65
CondInst	0.657	**0.217**	0.766	0.744	0.565	0.785	0.20
M2F + NDM*	0.924	1.000	**0.844**	0.843	**0.261**	0.881	0.85
M2F + OAEM*	0.907	1.109	0.823	0.866	0.370	0.882	0.80
Base (ours)	0.866	0.717	0.842	0.855	0.326	0.880	0.40
Gated stack (ours)	0.977	1.435	0.828	0.924	0.565	0.891	0.55
Stack + BPR (ours)	**0.983**	1.413	0.833	**0.936**	0.522	**0.902**	0.55
Offshore split (186 images, 374 GT ships)							
YOLACT	0.960	0.075	0.961	**0.992**	0.118	0.967	0.15
Mask R-CNN	0.933	0.081	0.946	0.928	0.048	0.951	0.75
Mask R-CNN + DCN	0.933	0.091	0.943	0.922	**0.038**	0.950	0.75
CondInst	0.928	**0.038**	0.953	0.976	0.081	0.968	0.10
M2F + NDM*	**0.987**	0.075	0.975	0.984	**0.038**	**0.983**	0.80
M2F + OAEM*	**0.987**	0.054	**0.980**	0.984	**0.038**	**0.983**	0.85
Base (ours)	0.963	0.167	0.941	0.936	0.075	0.949	0.40
Gated stack (ours)	**0.987**	0.086	0.972	0.979	**0.038**	0.980	0.50
Stack + BPR (ours)	**0.987**	0.086	0.972	0.979	**0.038**	0.980	0.50

$P_d = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$, so the missed-detection rate is $1 - P_d$; FA: false alarms per image; $F_1 = 2\mathrm{TP}/(2\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$; thr*: the F_1-optimal threshold, swept over $0.05 : 0.05 : 0.90$. Bold marks the best value in each column (P_d and F_1: higher is better; FA: lower is better); ties are both marked. Rows share the dataset split and matching evaluator; initialization, training budgets, inference computation, and checkpoint selection differ (Table VI). The optimal threshold is selected on the same evaluated split, and is therefore descriptive rather than independently validated. The stack and BPR rows precede learned re-ranking. *Our reimplementation on our carrier, not the original system (Table VI). P_d at a fixed threshold can be inflated by false alarms; read it together with FA, or use the F_1 columns.

2) Additional Frozen Carriers: Table VIII evaluates six additional carrier–dataset pairs covering five architectures, using one trained checkpoint per additional carrier and a separately fitted refiner. The fusion rule is reused without per-carrier retuning. Across the 66 displayed values, the 21 final split-wise endpoints exceed their respective first rows; the M2F block is contextual and uses the distinct rescoring operating point documented in its footnote. Inshore endpoint increments for the three HRSID carriers are 11.9/11.9/14.3, and for the three PSeg-SSDD carriers 6.9/16.3/6.1. The stack reduces PSeg-SSDD offshore AP by 0.6 for Mask R-CNN and 0.5 for YOLACT; their final increases are 1.4 and 2.7. Cascade R-CNN reaches 42.3 inshore, numerically 0.8 below the published 43.1 O2Former entry, but the training and inference protocols differ.

The observed stage increments vary by carrier. CondInst gains 8.9 inshore from the stack, versus 3.3/2.2 for the other PSeg-SSDD carriers. HRSID region-based carriers gain 8.8/8.7 from the stack and 3.1/3.2 from BPR; SOLOv2 gains 6.5 and 7.8, with a lower base AP_{75} of 15.7 versus 24.8/27.0. The proposed relation between initial AP and improvement is not statistically supported: Spearman correlation is -0.77 excluding CondInst ($n = 6$) and -0.11 including it ($n = 7$), neither significant. The selected images in Fig. 7 show Mask R-CNN detections increasing from 23 to 38 among 47 ships,

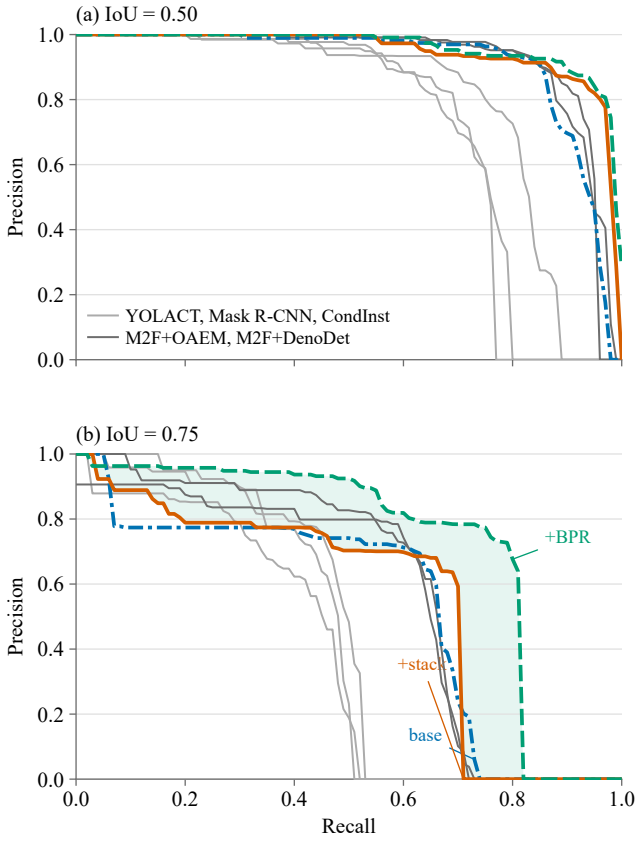

Fig. 4. Precision-recall curves on the PSeg-SSDD inshore split, from the stored predictions underlying Table VI before the learned re-ranking, at (a) IoU $= 0.50$ and (b) IoU $= 0.75$. Our three variants are labelled at the curve ends; the five competitors form generic (light gray) and SAR-specific (dark gray) tiers, named in (a)'s legend. The shaded band in (b) shows the difference between the stack and BPR curves; it is not an uncertainty interval. Per-method AP: Table VI. Offshore curves are omitted (saturated).

and M2F true positives from 31 to 71 among 88 ships while false alarms fall from 37 to 29.

G. Additional Negative Results

The retained negative results help delimit the tested design space. Despeckle-and-redraw and mild-Gamma variants score 37.1/37.3 inshore versus 38.8 for geometric TTA; seed-preserving variants outperform draw-majority fusion at 37.1 versus 27.0 and 37.3 versus 30.0. A verifier on the rejected score tail $[0.01, 0.35)$ gives 43.5/58.5/69.1 versus 43.6/58.6/69.1. The dumps retain scores down to zero but cap predictions at 100 per image, so this does not establish that every missed object has a recoverable candidate. Adding two decoder-depth descriptors to the 14-feature re-ranker gives 44.5/59.3/70.0 versus 44.5/59.4/70.1 in a post-hoc probe. A minimal-footprint carrier gives 28.2/46.3/60.5 versus 27.7/45.6/59.6 alone, but 36.4/51.6/63.0 versus 38.8/53.8/64.8 after stacking.

V. DISCUSSION

A. What the Decomposition Establishes

The most direct evidence in this study is the comparison of merge rules on a fixed candidate pool, followed by changes

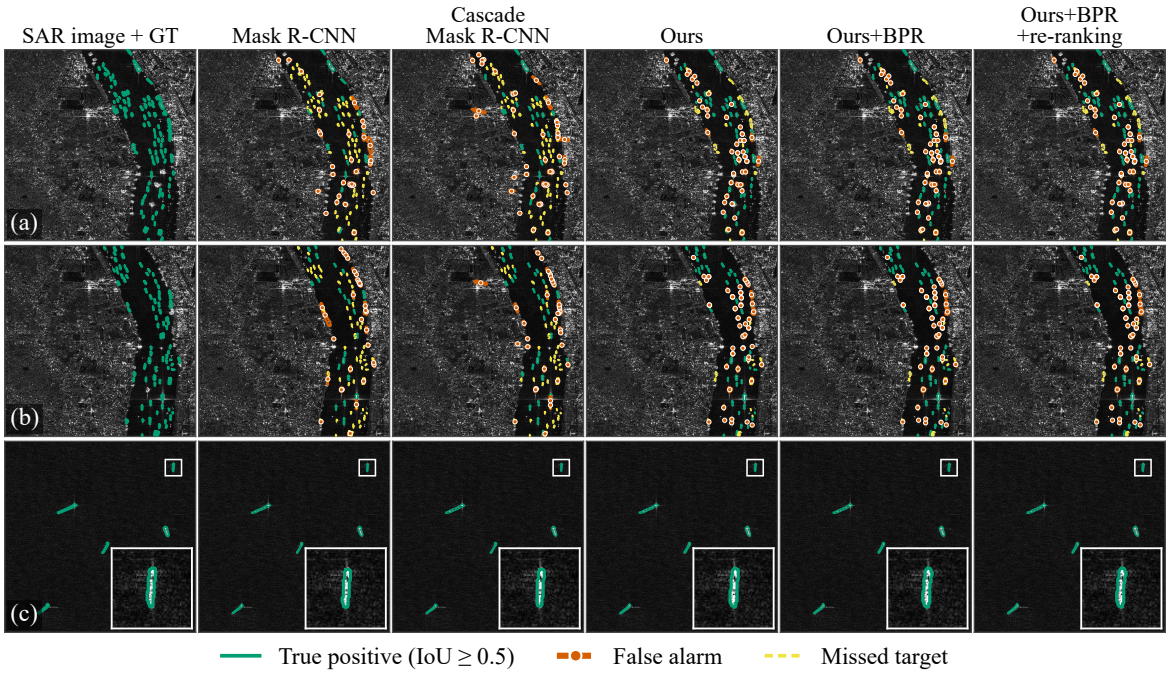

Fig. 5. Selected qualitative examples on three HRSID test scenes (0.3 score display threshold, mask IoU $\geq$ 0.5 matching). Green solid contours are true positives, orange dash-dot contours with a centroid dot false alarms, and yellow dashed contours missed targets. Columns: SAR image with ground truth, Mask R-CNN, Cascade Mask R-CNN, the training-free stack, + boundary-patch refiner, + learned re-ranker. Rows (a) and (b) are the inshore cases ranked most favorably for the cascade by visible-error reduction. Row (c) is an error-free offshore case selected for the largest inter-method mask-IoU spread. These are illustrative, not representative, samples.

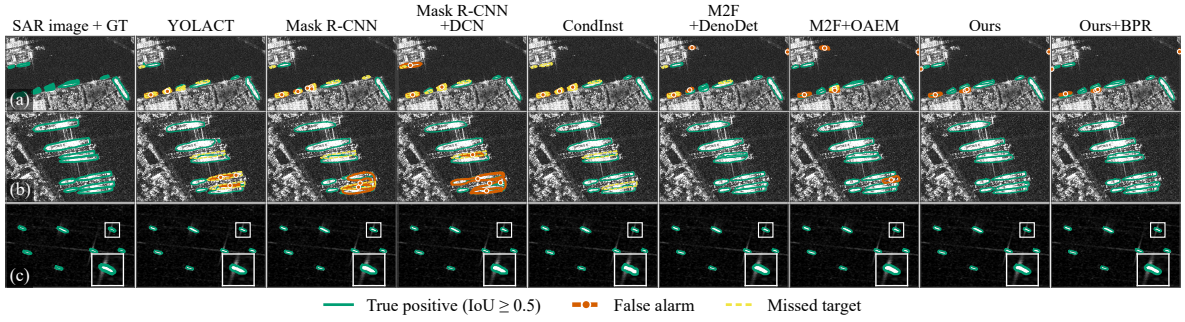

Fig. 6. Selected qualitative examples of the eight PSeg-SSDD methods of Table VI on three test scenes (0.3 score display threshold, mask IoU $\geq$ 0.5 matching; color coding as in Fig. 5; column names printed in the figure). The historical label "M2F+DenoDet denotes the local NDM reimplementation, and "M2F+OAEM denotes the local OAEM reimplementation; neither is the complete original system. Scenes follow the favorable-case selection rule of Section IV-E3. Rows (a) and (b) are inshore scenes; row (c) is offshore, where every method detects every ship and the inset magnifies the hull with the largest inter-method mask-IoU spread.

within the same trained carrier. Mask-space seed clustering accounts for part of the improvement over box-NMS; conditional voting adds a smaller inshore increment while retaining the seed-only offshore result among the originally tested rules. The matched per-view top-1 vote subsequently exceeds the gate on inshore, full, and offshore HRSID test partitions. Thus, the cluster-cardinality gate is not supported as a necessary algorithmic component by the current candidate-pool evidence. The stored BPR narrows the official top-1 advantage to 0.25 AP points and reverses the offshore ordering; this is an observational interaction on exposed test data, not a mechanism estimate. Re-ranking after top-1 has not been measured.

The write constraints make each component auditable, but do not imply commuting operators or guaranteed AP gains. Fusion changes the candidate set and its geometry; BPR changes mask IoU, and consequently the quality that a score should predict. Re-ranking changes the score order used by matching and top-ranked evaluation. Their interaction must therefore be assessed in the chosen sequence. The unchanged-pixel guarantee of BPR and unchanged-mask guarantee of re-ranking are implementation properties, distinct from claims about accuracy.

The architecture probes motivate hypotheses rather than a preservation law. For example, O2Former reports a 25.2-point inshore improvement over its 17.9 baseline under its own protocol [22]. Earlier local Gaussian–Wasserstein matching experiments report 12.5 to 16.9 (+4.4) with a from-scratch

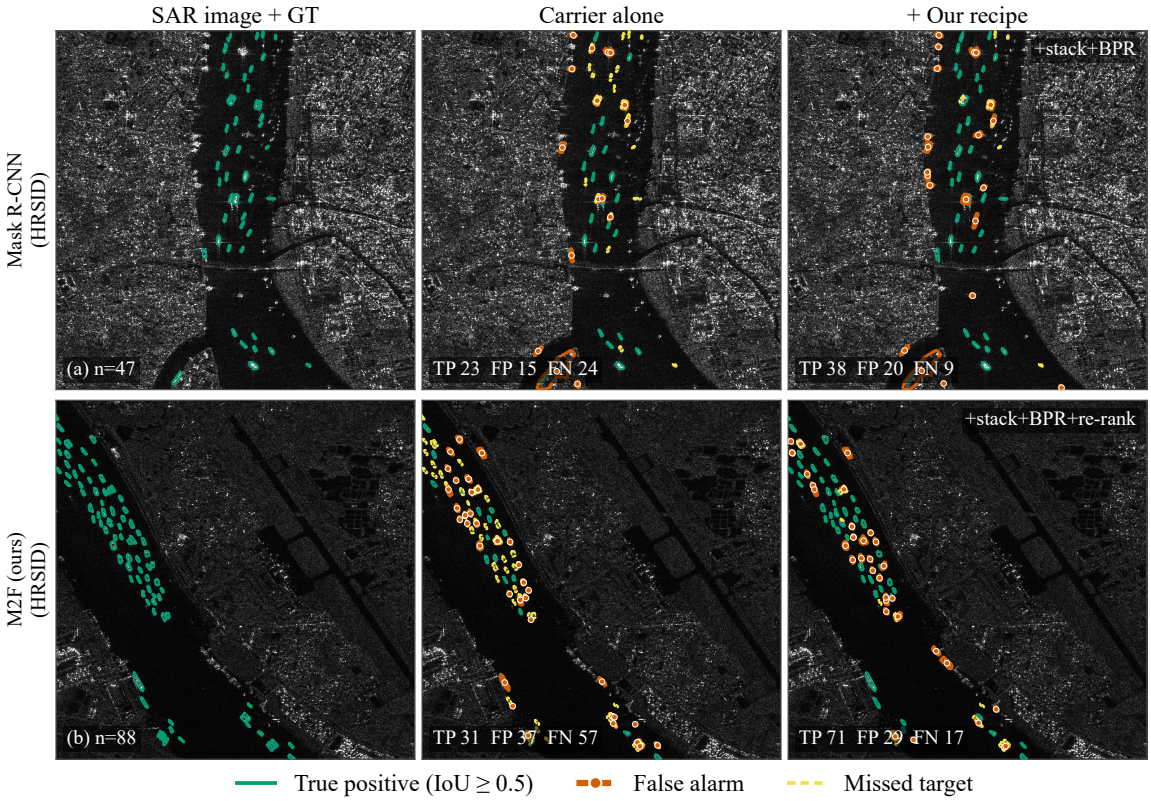

Fig. 7. Selected examples of the postprocessing recipe applied to two frozen carriers on the HRSID inshore split (Table VIII). Columns: SAR image with ground truth, carrier alone, carrier with the recipe applied (0.3 score display threshold; color coding as in Fig. 5; detection counts printed in each panel). Scenes were selected for visible improvement and do not estimate average behavior. Rows: (a) Mask R-CNN with the boundary-refined stack; (b) our Mask2Former carrier, which also applies the learned re-ranker. In row (b), carrier-only TP/FP counts are 31/37 and final TP/FP counts are 71/29.

head, comparing a single-seed base with a two-seed result, and 27.9 to 27.4 in a single-seed COCO-initialized comparison. These observations involve different training and aggregation conditions. Together with the 0.9-point SGA inshore loss, they suggest that training regime and operator design matter, but do not isolate pretraining, representation preservation, or capacity as causal explanations. In particular, the HRSID carrier itself received auxiliary mask-quality supervision, whereas the PSeg-SSDD carrier uses the standard head.

B. Limits and Confirmatory Evaluation

The principal limitation is selection. HRSID fusion settings were chosen on test results, the reported BPR instance is the best of three examined training seeds, and several comparison checkpoints were selected on test data. The bootstrap intervals condition on those decisions. The matched top-1 and reused-BPR results are also on the historically exposed HRSID test set and cannot be recycled as independent model-selection or generalization estimates. They remove two missing-control gaps while overturning the gate's claimed necessity; the historical complete cascade has not been compared with a top-1 cascade whose correctors are trained out of fold. A confirmatory study should allocate development images by parent SAR scene where possible, select the carrier, soup membership, readout, and correctors without access to the final evaluation set, and report pre-specified seeds and scene-level uncertainty. Repeating the recipe on PSeg-SSDD with a new carrier and refitted correctors provides exploratory evidence of portability under dataset-specific fitting, but its 46-image inshore subset yields intervals crossing zero for two stage increments.

Corrector supervision is another limitation. BPR's random patch holdout can share an image with its fitting patches; the re-ranker's image holdout consists of images already used to fit the carrier. Out-of-fold carrier predictions and image- or scene-disjoint corrector validation are needed to measure the effect of residual-distribution shift. Filename separation and annotation-free application do not settle parent-scene independence. Soup compatibility also needs a rule selected on development data: the six PSeg-SSDD seeds form two empirically distinct averaging groups, and increasing the member count is not consistently beneficial.

The current cost accounting includes three carrier trainings and separate corrector fits. View count alone does not determine latency, because input sizes, candidate counts, patch counts, and device transfers vary. A deployment comparison should measure synchronized mean and tail latency, memory, and per-stage work at fixed hardware, precision, and batching, alongside detection and mask quality. The limited amplitude-image datasets also leave cross-sensor and acquisition-domain transfer unresolved. A radiometric-consistency re-ranking probe returned +0.0 inshore over its flip-TTA control; that negative probe does not establish an information-theoretic sensor ceiling. The final HRSID offshore AP remains 70.1 versus the

TABLE VIII
TRANSFER OF THE POSTPROCESSING RECIPE ACROSS EVALUATED FROZEN CARRIERS (MASK AP $\times 100$)

Carrier	Variant	Inshore	Official	Offshore
HRSID				
Mask R-CNN	carrier alone	28.3	44.8	57.8
	+stack	37.1	52.2	63.6
	+BPR	40.2	56.2	68.2
Cascade R-CNN	carrier alone	30.4	46.5	59.2
	+stack	39.1	53.3	64.0
	+BPR	42.3	57.6	69.0
SOLOv2	carrier alone	19.0	38.2	54.0
	+stack	25.5	43.0	56.3
	+BPR	33.3	53.0	67.6
M2F (ours)	carrier alone	29.7	47.6	61.4
	+stack	38.8	53.8	64.8
	+BPR	43.6	58.6	69.1
	+re-rank	44.5	59.4	70.1
PSeg-SSDD				
Mask R-CNN	carrier alone	40.8	57.8	65.1
	+stack	44.1	58.6	64.5
	+BPR	47.7	61.3	66.5
CondInst	carrier alone	41.2	54.3	60.1
	+stack	50.1	59.7	63.6
	+BPR	57.5	66.1	69.3
YOLACT	carrier alone	38.9	55.7	63.4
	+stack	41.1	56.2	62.9
	+BPR	45.0	59.7	66.1

Each additional carrier uses one trained checkpoint; the M2F reference uses a three-member soup. "+stack" applies the wide eight-view multi-scale flip TTA with cluster-support-gated fusion, "+BPR" further applies a per-carrier boundary-patch refiner, and "+re-rank" further applies the learned re-ranker. Official is the full official test set of each dataset; inshore and offshore are its two scene subsets. The M2F carrier-alone row retains the previously reported soup result at λ=0.2 mask-IoU rescoring: 29.7/47.6/61.4, compared with the $\lambda = 0$ soup at 29.6/47.4/61.1 in Table I. It is contextual and is not an additional matched operating-point transfer test. Carrier fitting and checkpoint-selection protocols differ; only the fusion settings are reused without per-carrier retuning.

literature-reported 71.7, a 1.6-point difference, and the PSeg-SSDD differences remain 1.1 inshore and 2.1 offshore. These cross-protocol values provide context, not an estimate of an irreducible error floor.

VI. CONCLUSION

This study evaluated mask fusion, boundary-patch refinement, and quality-based score adjustment on trained SAR segmentation carriers. The selected gate uses candidate cardinality to choose between pixel voting and seed retention; the subsequent correctors constrain their writes to a morphological boundary band and detection scores, respectively. The selected historical cascade reports 44.5 inshore mask AP and 48.3 AP_{75} on HRSID, and the separately fitted PSeg-SSDD pipeline reports 62.1 inshore AP. The exact same-pool top-1 control exceeds the gate by 2.6 official mask AP points before BPR and by 0.25 points after the historically selected BPR, so the present evidence does not justify the gate as a necessary new method. Test-informed HRSID selection, in-sample corrector residuals, and parent-scene overlap prevent independent confirmation of the full cascade. PSeg-SSDD uses dataset-specific carrier and corrector fitting and does not establish frozen-model transfer or improvement from every stage. The next step is a newly locked source-disjoint study with strong simple readouts, out-of-fold residual supervision, and measured compute budgets.

DATA AND CODE AVAILABILITY

HRSID and PSeg-SSDD are used as described in the cited dataset and evaluation protocols. The implementation, configurations, prediction records, and checkpoints currently reside in a private local repository; no public code or checkpoint archive is established for this manuscript. The reported software environment uses MMDetection 3.3.0 and PyTorch 2.0.1. A reproducible public release would additionally require versioned split identifiers, checkpoint provenance, and the evaluation and selection records.

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OpenAI Codex was used to assist with restructuring and drafting Sections I–VI and with consistency checks against the supplied manuscript and implementation. OpenAI's built-in image-generation tool was used to generate the conceptual illustrations in Figs. 1, 2, and 3. These illustrations depict algorithmic operations; they do not constitute experimental data. The numerical results, recorded SAR examples, and precision–recall data originate from the existing research artifacts. Scientific validation and final approval of this working revision remain the responsibility of the authors.

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